

Lab Notebook: SimCLIP/FARE VLM Robustness Evaluation

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1. Motivation

Following the March 16 meeting, the project direction shifted from attack exploration to defense evaluation. Ahmed and George agreed the publishable contribution is demonstrating effective defenses against Geminio-style gradient inversion attacks.

Zarif Hossain shared the **Sim-CLIP** repository (<https://github.com/speedlab-git/SimCLIP>), which implements adversarial fine-tuning of CLIP's vision encoder. The question: **can adversarially robust VLMs defend against Geminio?**

Hypothesis

SimCLIP and FARE harden the CLIP vision encoder against adversarial input perturbations (PGD/APGD). If the vision encoder produces different (more robust) embeddings, this could affect the text-image similarity scores that Geminio uses to reshape the loss surface. If the attacker's VLM becomes less discriminative, the attack should weaken.

Threat Model Consideration

Important caveat identified before running experiments: **the threat models don't directly align**. In Geminio, the VLM (CLIP) runs on the **server/attacker side** to compute similarity scores. It is not exposed to adversarial perturbations — it processes real images. SimCLIP defends against adversarial *input* perturbations to trick CLIP into misclassifying. These are fundamentally different attack vectors. However, the experiments are still valuable because:

1. The hardened vision encoder produces different feature representations
 2. These different representations could incidentally affect similarity score distributions
 3. Negative results (no defense) would confirm the threat model mismatch and motivate FL-specific defenses
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2. Experimental Setup

2.1 VLM Variants Tested

Variant	Source	Architecture	Embedding Dim	Training Method
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BiomedCLIP	microsoft/BiomedCLIP-PubMedBERT_256-vit_base_patch16_224	ViT-B/16	512	Domain-specific (medical)
Vanilla CLIP	openai/clip-vit-large-patch14	ViT-L/14	768	Standard CLIP pretraining
SimCLIP-4	hossainzarif19/SimCLIP/simclip4.pt	ViT-L/14	768	Adversarial fine-tuning, symmetric cosine loss, $\text{eps} = 4/255$
FARE-4	chs20/fare4-clip	ViT-L/14	768	Adversarial fine-tuning, L2 loss, $\text{eps} = 4/255$

Note: SimCLIP and FARE only fine-tune the **vision encoder**. The text encoder is unchanged across all CLIP-based variants. This means text embeddings are identical — only image embeddings differ.

2.2 Verification

Before running experiments, we verified the adapter implementation:

- Text embeddings: cosine similarity between vanilla CLIP and SimCLIP-4 = **1.000000** (identical, as expected)
- Image embeddings: cosine similarity between vanilla CLIP and SimCLIP-4 = **0.9231** (differ meaningfully)
- Image embeddings: cosine similarity between vanilla CLIP and FARE-4 = **0.9143**

2.3 Domains

- **Medical:** ChestMNIST test set (22,433 images, 15 classes), ResNet18 backbone, 5 queries
- **UAV:** UAVScenes (4,029 images from AMtown01 + HKairport01, 18 classes), ResNet18 backbone, 6 queries

2.4 Training Configuration

All models trained identically to the original pipeline:

- Optimizer: Adam, $\text{lr} = 1\text{e-}3$
- Epochs: 5
- Temperature: $T = 100$
- Only classifier head trained (backbone frozen)
- Loss: Geminio loss reshaping: $\text{mean}(\text{per_sample_loss} * (1 - \text{softmax}(\text{clip_sim} * T)))$

2.5 Hardware

- DGX with 8x NVIDIA H200 (144 GB each)
 - Embedding computation: GPU 2
 - Model training: GPUs 1, 2, 3 (parallelized across VLM variants)
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3. Results

3.1 Medical Domain Loss Ratios (Top-10%)

Higher loss ratio = stronger attack (more gradient amplification on target images).

Query	BiomedCLIP (512d)	Vanilla CLIP (768d)	SimCLIP-4 (768d)	FARE-4 (768d)
Any chest X-ray showing pneumonia	10.83x	9.53x	14.40x	11.34x
Any chest X-ray showing cardiomegaly with enlarged heart	7.67x	7.14x	12.07x	9.55x
Any chest X-ray with pleural effusion	10.10x	6.77x	7.26x	15.30x
Any normal healthy chest X-ray	4.95x	5.71x	11.87x	10.07x
Any chest X-ray showing a lung mass or tumor	—	7.47x	11.70x	12.48x
Average	~8.39x	~7.32x	~11.46x	~11.75x
vs. Vanilla CLIP	+ 15%	baseline	+ 57%	+ 61%

3.2 UAV Domain Loss Ratios (Top-10%)

Query	Vanilla CLIP	SimCLIP-4	FARE-4
Swimming pool	126.98x	68.05x	50.40x
Solar panels	173.73x	45.05x	67.04x
Trucks on road	22.64x	82.80x	69.09x
River with bridge	153.51x	32.66x	33.40x
Airport runway	64.50x	208.12x	73.48x
Shipping containers	5.34x	44.54x	31.27x
Average	~91.12x	~80.20x	~54.11x
vs. Vanilla CLIP	baseline	-12%	-41%

3.3 Summary

Domain	Best for Attacker	Worst for Attacker	Defense Effectiveness
Medical	FARE-4 (11.75x)	Vanilla CLIP (7.32x)	None — robust VLMs help attacker
UAV	Vanilla CLIP (91.12x)	FARE-4 (54.11x)	Partial — FARE reduces 41% but still very high

4. Analysis

4.1 Why Robust VLMs Don't Defend Against Geminio

The results confirm the threat model mismatch hypothesis:

1. **Geminio doesn't attack the VLM.** The VLM processes clean, real images on the server side. Adversarial robustness of the vision encoder is irrelevant because no adversarial perturbations are applied to the inputs.
2. **Robust VLMs may produce *more* discriminative embeddings.** Adversarial training often forces the model to learn more semantically meaningful features (rather than texture shortcuts). More meaningful features = better text-image alignment = the attacker can more precisely identify target images. This explains why SimCLIP-4 and FARE-4 *increase* attack effectiveness in the medical domain.
3. **The medical domain benefits more from robust embeddings.** Medical images are visually similar (all chest X-rays). A VLM that captures more robust, semantically meaningful features can better distinguish pneumonia from cardiomegaly from normal — exactly what the attacker needs. The 57-61% improvement over vanilla CLIP is striking.
4. **UAV results are mixed because visual distinctiveness already saturates.** Vanilla CLIP already achieves 91x average loss ratio on UAV. The images are visually distinct enough that even a non-robust VLM can easily separate pools from runways. Robust VLMs don't add much here and may even lose some of the texture-level cues that helped vanilla CLIP.

4.2 Implications for Defense Strategy

This experiment demonstrates that **VLM-side hardening is orthogonal to gradient inversion defense**. The attack exploits gradient leakage in the FL protocol, not VLM vulnerabilities. Effective defenses must operate at the gradient level:

- **DP-SGD:** Add calibrated noise to gradients before sharing (with explicit epsilon budgets)
- **Gradient compression/pruning:** Reduce information content in shared gradients
- **Secure aggregation:** Cryptographic protocols preventing server from seeing individual updates
- **Federated FAML:** Ahmed/Zarif's robust FL framework — designed for FL-specific threats

This motivates the next phase of work: integrating Federated FAML as a defense and measuring its effectiveness against Geminio.

5. Artifacts Produced

5.1 Code

File	Purpose
prototype/vlm_simclip.py	Unified VLM adapter for vanilla CLIP, SimCLIP-4/2, FARE-4/2 via open_clip
prototype/ compute_embeddings.py	Unified Phase 1 embedding script for any domain + VLM combo

prototype/train_medical.py	Updated with --vlm flag (biomedclip, clip, simclip4, simclip2, fare4)
prototype/train_uav.py	Updated with --vlm flag (clip, simclip4, simclip2, fare4)
prototype/ dataset_medical.py	Updated with configurable embed_path parameter

5.2 Pretrained Weights

File	Size	Source
pretrained_vlm/ simclip4.pt	1.2 GB	https://huggingface.co/hossainzarif19/SimCLIP
pretrained_vlm/ simclip2.pt	1.2 GB	https://huggingface.co/hossainzarif19/SimCLIP
FARE-4	(loaded from hub)	hf-hub:chs20/fare4-clip

5.3 Computed Embeddings

File	Shape	VLM
data/medical-biomedclip-test.pt	[22433, 512]	BiomedCLIP (existing)
data/medical-clip-test.pt	[22433, 768]	Vanilla CLIP
data/medical-simclip4-test.pt	[22433, 768]	SimCLIP-4
data/medical-fare4-test.pt	[22433, 768]	FARE-4
uavscenes/uav_clip_embeddings_*.pt	[4029, 768]	Vanilla CLIP (existing)
uavscenes/uav_simclip4_embeddings_*.pt	[4029, 768]	SimCLIP-4
uavscenes/uav_fare4_embeddings_*.pt	[4029, 768]	FARE-4

5.4 Trained Malicious Models

15 new medical models in malicious_models_medical_v2/:

- 5 queries x 3 VLMs (clip, simclip4, fare4)

12 new UAV models in malicious_models_uav/:

- 6 queries x 2 VLMs (simclip4, fare4)

6. Reproducibility

Commands to reproduce

```
# Phase 1: Compute embeddings
python prototype/compute_embeddings.py --domain medical --vlm simclip4
python prototype/compute_embeddings.py --domain medical --vlm fare4 --g
python prototype/compute_embeddings.py --domain medical --vlm clip --gp
python prototype/compute_embeddings.py --domain uav --vlm simclip4 --gp
```

```
python prototype/compute_embeddings.py --domain uav --vlm fare4 --gpu 0

# Phase 2: Train malicious models
python prototype/train_medical.py --all --vlm simclip4 --output-dir ./m
python prototype/train_medical.py --all --vlm fare4 --output-dir ./mali
python prototype/train_medical.py --all --vlm clip --output-dir ./malic
python prototype/train_uav.py --all --vlm simclip4 --gpu 0
python prototype/train_uav.py --all --vlm fare4 --gpu 0
```

Environment

- Conda: geminio at /raid/scratch/dzimmerman2021/miniconda3/envs/geminio/
 - Python 3.9, PyTorch with CUDA 12.4
 - open_clip for SimCLIP/FARE/vanilla CLIP model loading
 - Hardware: NVIDIA H200 GPUs
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7. DP-SGD Defense Evaluation (April 1, 2026)

7.1 Motivation

Since VLM-side defenses failed (Section 4), we implemented DP-SGD — a gradient-level defense with formal privacy guarantees. DP-SGD (Abadi et al., CCS 2016) applies two steps to shared gradients:

1. **Gradient clipping:** bound the global L2 norm to $\max_grad_norm \ C$
2. **Gaussian noise:** add noise with $\sigma = C \times \sqrt{2 \ln(1.25/\delta)} / \epsilon$

This was implemented in `prototype/defenses.py` and integrated into both reconstruction scripts via `--dpsgd-epsilon` flag.

7.2 Configuration

- Clipping norm $C = 1.0$ (standard)
- $\delta = 1e-5$
- $\epsilon \in \{0.1, 1.0, 5.0, 10.0, 50.0\}$
- Applied post-hoc to shared gradients before gradient inversion
- 24,000 reconstruction iterations per experiment

7.3 Results — Medical (pneumonia_descriptive)

ϵ	LPIPS ↓	PSNR ↑	Attack F1	Privacy Level
None	1.019	10.86	0.625	No defense
0.1	0.788	11.35	0.375	Very strong
1.0	0.804	11.31	0.250	Strong
5.0	0.791	11.41	0.125	Moderate
10.0	0.797	11.28	0.250	Weak
50.0	0.836	11.63	0.625	Very weak

7.4 Results — UAV (swimming_pool)

ϵ	LPIPS ↓	PSNR ↑	Attack F1	Privacy Level
None	0.679	12.88	—	No defense
0.1	0.703	12.49	0.125	Very strong
1.0	0.693	12.50	0.125	Strong
5.0	0.689	12.39	0.000	Moderate
10.0	0.685	12.33	0.000	Weak
50.0	0.852	12.43	0.000	Very weak

7.5 Analysis

DP-SGD is the first defense that effectively stops Geminio in our experiments.

- Gradient clipping is the key mechanism.** Even $\epsilon = 50$ (minimal noise) stops UAV targeting completely ($F1 = 0.000$). The clipping bounds the gradient norm to $C = 1.0$, which directly disrupts Geminio's gradient amplification mechanism — the attack relies on target images producing disproportionately *large* gradients.
- UAV is more sensitive to DP-SGD than medical.** All UAV epsilon values achieve $F1 \leq 0.125$. Medical requires $\epsilon \leq 5$ to reach $F1 = 0.125$. This is the inverse of the undefended vulnerability (UAV was most vulnerable at 91x).
- Perceptual metrics improve or hold steady.** LPIPS actually decreases (improves) under DP-SGD for medical ($1.019 \rightarrow 0.788$). This suggests the defense disrupts *targeted* gradient amplification but overall reconstruction is comparable — the attack loses its ability to focus on specific images.
- Contrast with SimCLIP/FARE.** VLM robustness increased medical attack effectiveness by 57-61%. DP-SGD reduces it by 80% ($F1: 0.625 \rightarrow 0.125$). The defense operates at the right level — gradients, not embeddings.
- Non-monotonic medical F1.** The F1 doesn't decrease monotonically with ϵ (0.375 at $\epsilon = 0.1$, 0.125 at $\epsilon = 5$, 0.250 at $\epsilon = 10$). This likely reflects stochastic variation in the reconstruction — the clipping effect dominates over noise magnitude in this regime.

7.6 Artifacts

New code:

- `prototype/defenses.py` — shared defense module (pruning, noise, DP-SGD)
- Updated `prototype/reconstruct_medical.py` — `--dpsgd-epsilon` flag
- Updated `prototype/reconstruct_uav.py` — `--dpsgd-epsilon` flag

Results directories:

- `results/`
 `medical_pneumonia_descriptive_dpsgd_eps{0.1,1.0,5.0,10.0,50.0}/`
- `results/uav_swimming_pool_dpsgd_eps{0.1,1.0,5.0,10.0,50.0}/`

7.7 Reproducibility

```
# Medical DP-SGD experiments
for eps in 0.1 1.0 5.0 10.0 50.0; do
    python prototype/reconstruct_medical.py \
        --geminio-query pneumonia_descriptive \
        --dpsgd-epsilon $eps --gpu 0
done

# UAV DP-SGD experiments
for eps in 0.1 1.0 5.0 10.0 50.0; do
    python prototype/reconstruct_uav.py \
        --geminio-query swimming_pool \
        --dpsgd-epsilon $eps --gpu 0
done
```

8. FedAvg Multi-Epoch Defense (April 2, 2026)

8.1 Motivation

The Geminio paper (Section 4.3) showed that FedAvg with multiple local epochs weakens the attack because each SGD step modifies the model further from the attacker's crafted state, "washing out" malicious patterns. They tested this qualitatively but didn't evaluate it systematically with metrics.

8.2 Implementation

Added `simulate_fedavg()` to `prototype/defenses.py`. Instead of computing a single gradient (FedSGD), the client runs multiple local SGD steps on the received model and shares the parameter difference (`new_params - old_params`). Integrated via `--fedavg-epochs` and `--fedavg-lr` flags.

8.3 Results — Medical (pneumonia_descriptive, lr = 1e-3)

Local Epochs	LPIPS ↓	PSNR ↑	Attack F1
0 (FedSGD)	1.019	10.86	0.625
1	0.898	11.15	0.500
2	0.813	11.04	0.875
5	0.918	10.87	0.000
10	0.835	9.59	0.500
20	0.819	10.93	0.000

8.4 Results — UAV (swimming_pool, lr = 1e-3)

Local Epochs	LPIPS ↓	PSNR ↑	Attack F1
0 (FedSGD)	0.679	12.88	—
1	0.660	10.79	0.125
2	0.669	11.04	0.125
5	0.667	10.92	0.375

10	0.668	10.90	0.250
20	0.671	10.95	0.250

8.5 Analysis

1. **FedAvg provides some defense but is inconsistent.** Medical F1 is non-monotonic (0.875 at 2 epochs, 0.000 at 5, 0.500 at 10, 0.000 at 20). Stochastic variation in the 24K-iteration reconstruction dominates.
2. **UAV F1 stays low (0.125-0.375)** across all epoch counts — even 1 local epoch disrupts targeting. But it doesn't reliably reach 0.000 like DP-SGD.
3. **PSNR degrades for UAV** — drops from 12.88 (FedSGD) to ~ 10.9 across all FedAvg settings. Multiple local steps degrade overall reconstruction quality.
4. **DP-SGD is the stronger defense.** DP-SGD at $\epsilon = 5$ gives $F1 = 0.125$ (medical) and 0.000 (UAV) consistently. FedAvg is less predictable and has no formal privacy guarantees.
5. **Consistent with the paper's findings** — they showed FedAvg weakens the attack, especially with small learning rates. Our results confirm this but add the comparison showing DP-SGD is more reliable.

8.6 Complete Defense Comparison

Defense	Medical F1	UAV F1	Reliable?	Formal Guarantee?
None (FedSGD)	0.625	—	—	No
Gradient pruning 99%	inconsistent	inconsistent	No	No
Laplacian noise 1e-1	inconsistent	inconsistent	No	No
SimCLIP-4 (VLM)	— (worse)	— (mixed)	No	No
FARE-4 (VLM)	— (worse)	— (mixed)	No	No
FedAvg 5 epochs	0.000	0.375	Partial	No
FedAvg 20 epochs	0.000	0.250	Partial	No
DP-SGD $\epsilon = 5$	0.125	0.000	Yes	Yes (ϵ, δ)-DP

8.7 Artifacts

Updated code:

- `prototype/defenses.py` — added `simulate_fedavg()` function
- Updated `prototype/reconstruct_medical.py` — `--fedavg-epochs`, `--fedavg-lr` flags
- Updated `prototype/reconstruct_uav.py` — same flags

Results directories:

- `results/medical_pneumonia_descriptive_fedavg_e{1,2,5,10,20}_lr0.001/`
- `results/uav_swimming_pool_fedavg_e{1,2,5,10,20}_lr0.001/`

8.8 Reproducibility

```
# Medical FedAvg experiments
for e in 1 2 5 10 20; do
    python prototype/reconstruct_medical.py \
        --geminio-query pneumonia_descriptive \
        --fedavg-epochs $e --gpu 0
done

# UAV FedAvg experiments
for e in 1 2 5 10 20; do
    python prototype/reconstruct_uav.py \
        --geminio-query swimming_pool \
        --fedavg-epochs $e --gpu 0
done
```

9. Next Steps (Updated)

1. ~~DP-SGD evaluation~~ **DONE** — effective defense confirmed
2. ~~FedAvg multi-epoch evaluation~~ **DONE** — partial defense, less reliable than DP-SGD
3. ~~DP-SGD validation across queries~~ **DONE** — solar_panels confirms swimming_pool results
4. **Share results with Zarif** — SimCLIP/FARE negative result + DP-SGD/FedAvg positive results
5. **Integrate Federated FAML** — test Ahmed/Zarif's robust FL framework as complementary defense
6. **Combined defenses** — DP-SGD + FedAvg together, DP-SGD + pruning together
7. **Paper draft** — frame as: cross-domain attack, VLM robustness insufficient, DP-SGD effective, FedAvg analysis, Federated FAML